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# USE COMPUTER TECHNOLOGY TO RESEARCH A COMPUTER TOPIC

**UNIT STANDARD 114076**

**NQF LEVEL: 4**

**CREDITS: 3**

**NOTIONAL HOURS: 30**

| LEARNER INFORMATION | |
|---|---|
| **Name** | |
| **Surname** | |
| **ID Number** | |
| **Contact** | |

**LEARNER GUIDE**

## Learner Information:

| Details | Please Complete this Section |
|---|---|
| Name & Surname: | |
| Organisation: | |
| Unit/Dept: | |
| Facilitator Name: | |
| Date Started: | |
| Date of Completion: | |

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|---|---|
| **Unit Std #** | 114076 |
| **NQF Level** | 4 |
| **Notional hours** | 30 |

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|---|---|
| **Credit(s)** | 3 |
| **Field** | Field 03 - Physical, Mathematical, Computer and Life Sciences |
| **Sub-Field** | Information Technology and Computer Sciences |
| **Qualification** | NATIONAL CERTIFICATE: INFORMATION TECHNOLOGY: SYSTEMS SUPPORT: SAQA 48573 -LEVEL 5- 147 CREDITS |

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| **Specific Outcome 1 :Plan the research of a computer topic.** |
| • The plan identifies the topic, objectives, and scope of the research.<br>• The plan identifies the time to be taken for the research, the research methods to be used, and the sources of information to be used.<br>• The plan identifies the target audience, presentation methods, and the computer applications to be used for the analysis of data and the presentation of the results of the research. |
| **Specific Outcome 2 : Conduct research of a computer topic using computer technology** |
| • The research conducted accumulates data according to the research plan.<br>• The research conducted provides data analysis with conclusions.<br>• The description of the analysis methods allows the validity of the analysis to be assessed.<br>• Research progress is indicated at intervals by reports, according to the research plan.<br>• The research conducted uses a computer application to analyse the research data. |
| **Specific Outcome 3 : Present the results of research of a computer topic using computer technology** |

- The presentation is made using the computer application identified in the research plan.
- The presentation communicates summarised research data and conclusions to the target audience.

**NOTIONAL HOURS BREAKDOWN**

The candidate undertaking this unit standard is best advised to at least spend one hundred hours of study time on this learning programme. Below is a table which demonstrates how these one hundred hours could be spread:

**TIMEFRAME**

| **Total Notional Hours** | **Contact Time** | **Non contact- Self-Study/Assessment** |
|---|---|---|
| Credits (3) x 10 = 30 | 9HRS | 21HRS |

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **1.** | **Learning Programme Name:** | | REFER TO COVER PAGE | | | | | | | | |
| **2.** | **SAQA Qualification/Unit Standard Title:** | | REFER TO COVER PAGE | | | | | | | | |
| **3.** | **Qualification/ Unit Standard** | | **4.** | **SAQA ID Number** | | **5.** | **NQF Level** | 4 | **6.** | **Credits** | **3** |
| **7.** | **<u>PURPOSE</u> for offering this programme to your learners:** | | REFER TO NEXT PAGE | | | | | | | | |

| | | |
|---|---|---|
| **8.** | **<u>TARGET AUDIENCE</u> for this specific programme:** | REFER TO NEXT PAGE |
| **9.** | **Entry/Admission Requirements:** | REFER TO NEXT PAGE |
| **10.** | **Timeframe for Training: (Total Hours/Days/Weeks)** | **Theory content** –Role play, Simulation, Group work, Pair work = 9 hrs.<br>**Non contact session**- self-study, assignment, practise guided by coach or mentor, formative assessment and summative assessment =21 hrs. |

# The Learner guide

**At the end of this unit standard you will be able toUse computer technology to research a computer topic**

| **Purpose:** |
| --- |
| **People credited with this unit standard are able to:**<br><br>• To provide a competent knowledge of the areas covered<br>• For those working in, or entering the workplace in the area of End-User Computing<br>• As additional knowledge for those wanting to understand the areas covered |

| **Specific outcome:** |
| --- |
| • Plan the research of a computer topic<br>• Conduct the research of a computer topic using computer technology, and to<br>• Present the results of research of a computer topic using computer technology. |

| **Learning assumed to be in place:** |
| --- |
| • Operate a personal computer<br>• Produce presentation documents for business<br>• Demonstrate ability to use the World Wide Web. |

| **Equipment needed:** |
| --- |
| **Learning material, Learner workbook, Pen, Ruler.** |
| PLEASE NOTE: THE USE OF PENCILS OR TIPPEX IS NOT ALLOWED.<br>IF YOU USE A PENCIL THE VALIDITY OF YOUR WORK COULD BE QUESTIONABLE, AND THIS COULD LEAD TO FRAUD. |

| Resources (selective resources might be used, depending on the facilitator and venue circumstances), one or all of the following can be used: |
|---|
| • Your facilitator/mentor<br>• Learning material<br>• Learner workbook<br>• Visual aids<br>• White board<br>• Flip chart<br>• Equipment<br>• Training venue |

| **Venue, Date and Time:** |
|---|
| Consult your facilitator should there be any changes to the venue, date and/or time.<br>Refer to your timetable. |

| **Assessments:** |
|---|
| **The only way to establish whether you are competent and have accomplished the specific outcomes is through continuous assessments**<br>**The given exercises can contain one or more of the following:**<br>• Information for you to read<br>• Exercises that require you to have a problem-solving approach to communication<br>• Questions for you to answer<br>• Case studies with questions that follow<br><br>**How to do the exercise:**<br><br>• The facilitator will tell you which exercise you need to complete each day.<br>• You need to hand in your answers to the facilitator who will mark it for correctness.<br>• If you do not know the answer, you will have to go back to that particular section in you learner guide and go over it again.<br>• Ask the facilitator for help, if you do not understand any of the questions asked.<br>• Always remember to give reasons for your answers |

## SPECIFIC OUTCOME 1:

## Plan the research of a computer topic.

## ASSESEMENT CRITERIA

- The plan identifies the topic, objectives, and scope of the research.
- The plan identifies the time to be taken for the research, the research methods to be used, and the sources of information to be used.
- The plan identifies the target audience, presentation methods, and the computer applications to be used for the analysis of data and the presentation of the results of the research.

### 1.1 Planning steps of research

**STEP 1: Identify the problem or topic**

Identify a research problem or area of interest from everyday life experiences, practical issues, past research, or theory. Pay attention to the feasibility of your research problem or topic and whether it can be researched systematically. Determine the resources needed to conduct the study, your interest level, its size and complexity, as well as the value of your results or solution for both theory and practice.

To thoroughly describe the research problem or topic, create a statement that includes the educational topic or specific problem and the justification for research.

**STEP 2: Review prior research**

Explore the research literature to gain an understanding of the current state of knowledge pertaining to your research problem. A review of prior research will inform you if your research problem has already been explored (and if a revision or replication is needed), how to design your study, what data collection methods to use, and how to make sense of the findings of your study once data analysis is complete. Reviewing prior research can also help with creating research questions, what population to explore, and laying the theoretical groundwork for your study.

If you are conducting qualitative research, this step is sometimes used throughout the research process or after data is collected (e.g., grounded theory research).

**STEP 3: Determine the Research Purpose, Research Questions, or Hypotheses**

Identifying a clear purpose and creating a purpose statement helps determine how the research should be conducted, what research design to use, and the research question(s) or hypothesis(es) of your study. Four general purposes for conducting educational research are to explore, describe, predict, or explain the relation between two or more educational variables.

**Explore** – an attempt to generate ideas about educational phenomenon
**Describe –** an attempt to describe the characteristics of educational phenomenon

**Predict** – an attempt to forecast an educational phenomenon

**Explain** – an attempt to show why and how an educational phenomenon operates

The purpose of your study will help you determine which research design you should follow

## 1.2 Introduction to computer research

Qualitative research methods are being used increasingly in evaluation studies, including evaluations of computer systems and information technology. This chapter provides an overview of the nature and appropriate uses of qualitative methods and of key considerations in conducting qualitative research.

The goal of qualitative research is understanding issues or particular situations by investigating the perspectives and behavior of the people in these situations and the context within which they act. To accomplish this, qualitative research is conducted in natural settings and uses data in the form of words rather than numbers. Qualitative data are gathered primarily from observations, interviews, and documents, and are analyzed by a variety of systematic techniques. This approach is useful in understanding causal processes, and in facilitating action based on the research results.

Qualitative methods are primarily inductive. Hypotheses are developed during the study so as to take into account what is being learned about the setting and the people in it. Qualitative methods may be combined with quantitative methods in conducting a study. Validity threats are addressed primarily during data collection and analysis.

**Reasons for Qualitative Research**

The reasons for using qualitative methods in evaluating computer information systems:

1. Understanding how a system's users perceive and evaluate that system and what meanings the system has for them Users' perspectives generally are not known in advance.

It is difficult to ascertain or understand these through purely quantitative approaches. By allowing researchers to investigate users' perspectives in depth, qualitative methods can contribute to the explanation of users' behavior with respect to the system, and thus to the system's successes and failures and even of what is considered a "success" or "failure".

2. Understanding the influence of social and organizational context on systems use Computer information systems do not exist in a vacuum; their implementation, use, and success or failure occur in a social and organizational context that shapes what happens when that system is introduced. Some researchers consider this so important as to treat "context" as intrinsically part of the object of study rather than as external to the information system. Because of "context," in important respects, a system is not the same system when it is introduced into different settings. As is true for users' perspectives, the researcher usually does not know in advance what all the important contextual influences are. Qualitative methods are useful for discovering and understanding these influences, and also for developing testable hypotheses and theories.

3. Providing formative evaluation that is aimed at improving a program under development, rather than assessing an existing one. Although quantitative and experimental designs often are valuable in assessing outcomes, they are less helpful in giving those responsible for systems design and implementation timely feedback on their actions. Qualitative evaluation can help both in system design as well as in studies of system use

**Research Questions and Evaluation Goals**

Qualitative methods typically are used to understand the perception of an information system by its users, the context within which the system is implemented or developed, and the processes by which changes occur or outcomes are generated. They usually focus on the description, interpretation, and explanation of events, situations, processes, and outcomes, rather than the correlation of variables, and tend to be used for understanding a particular case or for comparison of a small number of cases, rather than for generalization to a specified population. They are useful for systematically collecting so-called "anecdotal" evidence and turning the experiences they describe into data that can be rigorously collected and analyzed.

Thus, the questions posed in a qualitative study are initially framed as "what," "how," and "why" queries, rather than as whether a particular hypothesis is true or false. The

fundamental question is "What is going on here?" This question is progressively narrowed, focused, and made more detailed as the evaluation proceeds. Qualitative studies may begin with specific concerns or even suppositions about what is going on, but major strengths of qualitative methods are avoiding tunnel vision, seeing the un- expected, disconfirming one's assumptions, and discovering new ways of making sense of what is going on. Qualitative evaluators typically begin with questions such as:

- What is happening here?
- Why is it happening?
- How has it come to happen in this particular way?
- What do the people involved think is happening?
- How are these people responding to what is happening?
- Why are these people responding that way?

### 1.3 Other considerations to make in planning a research project

**Designing**

The design stage constitutes the step where the methodological procedure is planned and prepared. What is the time schedule and how do the different steps interrelate? When the chosen technique is interviews, designing the research project will be to determine which kind of interviews to use—personal, collective (focus group), expert, etc.—and how many interviews to perform.

**Reporting**

It goes without saying that reporting covers the part of the research process where the researcher writes a report to present his findings.

## SPECIFIC OUTCOME 2 :

## Conduct research of a computer topic using computer

| technology. |
| --- |
| **ASSESEMENT CRITERIA** |
| ❖ The research conducted accumulates data according to the research plan.<br>❖ The research conducted provides data analysis with conclusions.<br>❖ The description of the analysis methods allows the validity of the analysis to be assessed.<br>❖ Research progress is indicated at intervals by reports, according to the research plan.<br>❖ The research conducted uses a computer application to analyse the research data. |

## 2.0 What do we mean by collecting data?

Essentially, collecting data means putting your design for collecting information into operation. You've decided how you're going to get information – whether by direct observation, interviews, surveys, experiments and testing, or other methods – and now you and/or other observers have to implement your plan. There's a bit more to collecting data, however. If you are conducting observations, for example, you'll have to define what you're observing and arrange to make observations at the right times, so you actually observe what you need to. You'll have to record the observations in appropriate ways and organize them so they're optimally useful.

Recording and organizing data may take different forms, depending on the kind of information you're collecting. The way you collect your data should relate to how you're planning to analyze and use it. Regardless of what method you decide to use, recording should be done concurrent with data collection if possible, or soon afterwards, so that nothing gets lost and memory doesn't fade.

**Some of the things you might do with the information you collect include:**

- Gathering together information from all sources and observations
- Making photocopies of all recording forms, records, audio or video recordings, and any other collected materials, to guard against loss, accidental erasure, or other problems
- Entering narratives, numbers, and other information into a computer program, where they can be arranged and/or worked on in various ways
- Performing any mathematical or similar operations needed to get quantitative information ready for analysis. These might, for instance, include entering numerical

observations into a chart, table, or spread sheet, or figuring the mean (average), median (midpoint), and/or mode (most frequently occurring) of a set of numbers.

- Transcribing (making an exact, word-for-word text version of) the contents of audio or video recordings
- Coding data (translating data, particularly qualitative data that isn't expressed in numbers, into a form that allows it to be processed by a specific software program or subjected to statistical analysis)
- Organizing data in ways that make them easier to work with. How you do this will depend on your research design and your evaluation questions. You might group observations by the dependent variable (indicator of success) they relate to, by individuals or groups of participants, by time, by activity, etc. You might also want to group observations in several different ways, so that you can study interactions among different variables.

## 2.1 Ethics for gathering data

Consider several ethical issues related to professionalism, and the collection and storing of data from human subjects when conducting educational research.

**Professional issues**

One important issue for researchers is to present truthful results. There is never any justification for misrepresentation or fraud, and the cost is enormous, both to the researcher and professional community.

**Human subject research**

Many professional organizations have prepared ethical guideless for educational research with human subjects such as the Department of Health, Education, and Welfare

## 2.2 Data Collection in computer science

The most important principle of qualitative data collection is that every- thing is potential data. The evaluator does not rigidly restrict the scope ofdata collection in advance, nor

use formal rules to decide that some data are inadmissible or irrelevant. However, this approach creates two potential problems: validity and data overload.

**Validity** issues are addressed below. The problem of data overload is in some ways more intractable. The evaluator must continually make decisions about what data are relevant and may change these decisions over the course of the project. The evaluator must work to focus the data collection process, but not to focus it so narrowly as to miss or ignore data that would contribute important insights or evidence.

Qualitative evaluators use three main sources for data: (1) observation, (2) open-ended interviews and survey questions, and documents and texts. Qualitative studies generally collect data by using several of these methods to give a wider range of coverage. Data collection almost always involves the researcher's direct engagement in the setting studied, what often is called "fieldwork." Thus, the researcher is the instrument for collecting andanalyzing data; the researcher's impressions, observations, thoughts, and ideas also are data sources. The researcher incorporates these when recording qualitative data in detailed, often verbatim form as field notes or interview transcripts. Such detail is essential for the types of analy- sis that are used in qualitative research. We discuss each of these data sources in turn, drawing again on Kaplan and Duchon's study and several other studies for examples.

**Observation**

Observation in qualitative studies typically involves the observer's active involvement in the setting studied; it is usually called "participant observation" to distinguish it from passive or non-interactive observation. Participant observation allows the observer to ask questions for clarification of what is taking place and to engage in informal discussion with system users, as well as to record on-going activities and descriptions of the setting. It produces detailed descriptive accounts of what was going on (including verbal interaction), as well as eliciting the system users' own explanations, evaluations, and perspectives in the immediate context of use, rather than retrospectively. Such observation often is crucial to the assessment of a system

**Open-Ended Interviews and Survey Questions**

Open-ended interviewing requires a skilful and systematic approach to questioning participants. This can range frominformal and conversational interviews to ones with a specific agenda. There are two distinctive feature of open-ended interviewing. First, the

goal is to elicit the respondent's views and experiences in his or her own terms, rather than to collect data that are simply a choice among pre-established response categories. Second, the interviewer is not bound to a rigid interview format or set of questions, but should elaborate on what is being asked if a question is not understood, follow up on unanticipated and potentially valuable information with additional questions, and probe for further explanation.

Another way to investigate the views of groups of respondents is through focus groups. This involves interviewing several people together, and adds an opportunity for those present to react and respond to each others' remarks

**Documents and Texts**

Documents, texts, pictures or photographs, and artifacts also can be valuable sources of qualitative data.

## 2.3Summary of data collection methods in research

**Methods**

From the list below, select the data gathering method(s) you wish to use and learn the best practices way to implement it.

**Content analysis**

The systematic examination of oral, written, or visual communication.

**Experiment**

A variety of research designs that use before and after, and/or group comparisons, to measure cause and effect relations.

**Focus group**

A group of similar individuals who provide information during a directed and moderated interactive group discussion.

**Interview**

A directed conversation with an individual using a list of questions designed to gather extended responses.

**Observation**

The systematic observation of behavior using checklists, scaled ratings, or narrative comments.

**Survey**

An ordered series of questions administered to individuals in a systematic manner.

### 2.4 When and by whom should data be collected and analyzed?

- You can hire or find a volunteer outside evaluator, such as from a nearby college or university, to take care of data collection and/or analysis for you.
- You can conduct a less formal evaluation. Your results may not be as sophisticated as if you subjected them to rigorous statistical procedures, but they can still tell you a lot about your program. Just the numbers – the number of dropouts (and when most dropped out), for instance, or the characteristics of the people you serve – can give you important and usable information.
- You can try to learn enough about statistics and statistical software to conduct a formal evaluation yourself. (Take a course, for example.)
- You can collect the data and then send it off to someone – a university program, a friendly statistician or researcher, or someone you hire – to process it for you.
- You can collect and rely largely on qualitative data. Whether this is an option depends to a large extent on what your program is about. You wouldn't want to conduct a formal evaluation of effectiveness of a new medication using only qualitative data, but you might be able to draw some reasonable conclusions about use or compliance patterns from qualitative information.
- If possible, use a randomized or closely matched control group for comparison

## SPECIFIC OUTCOME 3:

## Present the results of research of a computer topic using

| computer technology |
| --- |
| **ASSESEMENT CRITERIA** |
| ❖ The presentation is made using the computer application identified in the research plan.<br>❖ The presentation communicates summarised research data and conclusions to the target audience. |

### 3.1 Data Analysis using computer technology

The basic goal of qualitative data analysis is understanding: the search for coherence and order. The purpose of data analysis is to develop an under- standing or interpretation that answers the basic question of what is going on here. This is done through an iterative process that starts by developing an initial understanding of the setting and perspectives of the people being studied. That understanding then is tested and modified through cycles of additional data collection and analysis until an adequately coherent interpretation is reached Thus, in qualitative research, data analysis is an ongoing activity that should start as soon as the project begins and continue through the entire course of the research [5]. The processes of data collection, data analysis, interpretation, and even research design are intertwined and depend on each other.

**We briefly discuss each of the four techniques.**

**Coding**

The purpose of coding, in qualitative research, is different from that in experimental or survey research or content analysis. Instead of applying a pre-established set of categories to the data according to explicit, unambiguous rules, with the primary goal being to generate frequency counts of the items in each category, it instead involves selecting particular segments of data and sorting these into categories that facilitate insight, comparison, and the development of theory [46]. While some coding categories may be drawn from the evaluation questions, existing theory, or prior knowledge of the setting and system, others are developed inductively by the evaluator during the analysis, and still others are taken from the language and conceptual structure of the people studied. The key feature of most qualitative coding is that it is grounded in the data (i.e., it is developed in interaction with, and is tailored to the understanding of, the particular data being analyzed).

**Analytical Memos**

An analytical memo is anything that a researcher writes in relationship to the research, other than direct field notes or transcription. It can range from a brief marginal comment on a transcript, or a theoretical idea incorporated into field notes, to a full-fledged analytical essay. All of these are ways of getting ideas down on paper, and of using writing as a way to facilitate reflection and analytical insight. Memos are a way to convert the researcher's perceptions and thoughts into a visible form that allows reflection and further manipulation. Writing memos is an important analysis technique, as well as being valuable for many other purposes in the research, and should begin early in the study, perhaps even before starting the study.

**Displays**

Displays, such as matrices, flowcharts, and concept maps, are similar to memos in that they make ideas, data, and analysis visible and permanent. They also serve two other key functions: data reduction, and the presentation of data or analysis in a form that allows it to be grasped as a whole. These analytical tools have been given their most detailed elaboration by Miles and Huberman, but are employed less self-consciously by many other researchers. Such displays can be primarily conceptual, as a way of developing theory, or they can be primarily data oriented. Data-oriented displays, such as matrices, can be used as an elaboration of coding; the coding categories are presented in a single display in conjunction with a reduced subset of the data in each category. Other types of displays, such as concept maps, flowcharts, causal networks, and organizational diagrams, display connections among categories.

**Contextual and Narrative Analysis**

Contextual and narrative analysis has developed mainly as an alternative to coding. Instead of segmenting the data into discrete elements and resorting these into categories, these approaches to analysis seek to understand the relationships between elements in a particular text, situation, or sequence of events. Methods such as discourse analysis , narrative analysis , conversation analysis ; profiles, or ethnographic microanalysis identify therelationships among the different elements in that particular interview or situation, and their meanings for the persons involved, rather than aggregating data across contexts.

**Software**

Qualitative methods produce large amounts of data that may not be readily amenable to manipulation, analysis, or data reduction by hand. Computer software is available that

can facilitate the process of qualitative analysis. Such programs perform some of the mechanical tasks of storing and coding data, retrieving and aggregating previously coded data, and making connections among coding categories, but do not "analyze" the data in the sense that statistical software does. All of the conceptual and analytical work of making sense of the data still needs to be done by the evaluator. There are different types of programs, some developed specifically for data analysis, and others (including word processors, textbase managers, and network builders) that can be used for some of the tasks of analysis. For relatively small-scale projects, some qualitative researchers advocate not using any software besides a good word processor. A very sophisticated and powerful program may be difficult to use if it has unneeded features, so it is advisable to carefully consider what the program needs to do before committing to its use.

**Validity**

Validity in qualitative research addresses the necessarily "subjective" nature of data collection and analysis. Because the researcher is the instrument for collecting and analyzing data, the study is subjective in the sense of being different for different researchers. Different researchers may approach the same research question by collecting different data or by interpreting the same data differently.

Qualitative researchers acknowledge their role as research instruments by making it an explicit part of data collection, analysis, and reporting. As in collecting and analyzing any data, what the evaluator brings to the task— his or her biases, interests, perceptions, observations, knowledge, and critical faculties—all play a role in the study. Qualitative researchers include in their studies specific ways to under- stand and control the effects of their background and role. They recognize that the relationships they develop with those studied have a major effect on the data that can be gathered and the interpretations that can be developed.

**Rich Data**

Rich data are data that are detailed and varied enough that they provide a full and revealing picture of what is going on, and of the processes involved. Collecting rich data makes it difficult for the researcher to see only what supports his or her prejudices and expectations and thus provides a test of one's developing theories, as well as provides a basis for generating, developing, and supporting such theories.

**Feedback or Member Checking**

This is the single most important way of ruling out the possibility of misinterpreting the meaning of what participants say and do or what the researcher observed, and the perspective the participants have on what is going on. Feedback, or member checking, involves systematically gathering feedback about one's conclusions from participants in the setting studied and from others familiar with the setting. The researcher checks that the interpretation makes sense to those who know the setting especially well. In addition, this is an important way of identifying the researcher's biases and affords the possibility for collecting additional important data

**Searching for Discrepant Evidence andNegative Cases**

Identifying and analyzing discrepant data and negative cases is a key part of the logic of validity testing in qualitative research. Instances that cannot be accounted for by a particular interpretation or explanation can point up important defects in that account. There are strong pressures to ignore data that do not fit prior theories or conclusions, and it is important to rigorously examine both supporting and discrepant data. In particularly difficult cases, the only solution may be to report the discrepant evidence and allow readers to draw their own conclusions

### 3.2.1 Reporting observation results

**Analyze the data**

Analyze the observational data by reviewing the completed observation form and any written comments. You should also have a face-to-face debriefing discussion with all observers if you did not conduct all observations yourself. This will give the observers an opportunity to explain the data and provide additional feedback. Begin reviewing the data after the first observation. Look for patterns in the data using the research questions or hypotheses of your study to focus analysis. Other questions or issues may emerge as you discuss and review the information.

**Determine findings**

Summarize findings based on your analysis and in relation to your research questions and previous research findings. Verify that findings are grounded in what was observed.

**Report results**

How you report your findings depends on the types of observations you used. If you used qualitative observations (e.g., narrative comments), present repeating ideas that lead to major themes that, in turn, inform conclusions and implications. Provide one or two

examples of a repeating idea. You may also want to note an exception to a trend in order to highlight a noteworthy idea.

For quantitative observations (e.g., rating scales), present the statistical findings graphically and with the level of detail useful to the audience. For all observation types, make sure to discuss what practical or theoretical implications can be drawn for the findings, any major shortcomings or limitations of the methodology used, and directions or suggestions for future research.

### 3.2.2 Reporting interview results

**Transcribe and analyze the data**

Interviews generate large quantities of data and tape transcription typically takes four to six hours for each hour of speech, although using a transcription machine or having good typing skills can reduce the time. It is important, therefore, to have a clear plan to guide this phase of the study. In addition, condensing, organizing, and making meaning of interviews is often the most time-consuming and expensive part of analysis.

**Determine findings**

View analyzed data from a distance until you see a larger picture and understand how this picture relates to your research question(s). Similar research may help you make sense of repeating ideas and larger themes. For example, you might identify underlying factors that explain the themes you have observed and then construct a logical chain of evidence. You might also describe an adaptive or maladaptive process that captures the behavior of respondents. If there are respondents who do not follow the usual pattern, it may be important to understand why. Qualitative researchers need to be flexible and open to the unexpected. Drawing on repeating ideas and themes, summarize the findings in relation to your research question(s) and to previous research.

When interpreting qualitative data, verify your findings. Review your data repeatedly to check that your findings are grounded in what was said. Look at independent evidence from other sources and use other methods, such as surveys, focus groups, or experiments, to triangulate your findings. To improve the study's reliability and validity , show your results to some of the interviewees and ask them if you have accurately recorded what they meant.

**Report results**

To report qualitative results, present repeating ideas that lead to major themes that, in turn, inform conclusions and implications. Conclusions are statements that interpret and evaluate the results found from the study. Make sure to give primary emphasis to the results that relate to the research questions of your study.

Quote one or two responses that exemplify a repeating idea. Quotations, which capture the words, emotions, experiences, and perceptions of interviewees, are not easily dismissed by readers. You may also want to quote a response that was an exception to a trend in order to illustrate a minority opinion or highlight a noteworthy idea. If so, report that it is one person's response.Finally, make sure to discuss what practical or theoretical implications can be drawn for your findings, any major shortcomings or limitations of the methodology used, and directions or suggestions for future research

### 3.2.3 Reporting focus group results

**Transcribe and analyze the data**

Data analysis may be relatively simple, involving a summary of major themes, or may call for more complex content analyses and comparisons of groups (Goldenkoff, 2004). A brief summary and analysis, highlighting major themes, is sufficient when the results are readily apparent or the purpose of the focus group is supplemental. On the other hand, to get an in-depth understanding of a complex issue you should conduct a systematic analysis using full transcripts [more] and a formalized coding scheme.

**Determine findings**

View analyzed data from a distance until you see a larger picture and understand how this picture relates to your research question(s). Similar research may help you make sense of repeating ideas and larger themes. For example, you might identify underlying factors that explain the themes you have observed and then construct a logical chain of evidence. You might also describe an adaptive or maladaptive process that captures the behavior of respondents. If there are respondents who do not follow the usual pattern, it may be important to understand why. Qualitative researchers need to be flexible and open to the unexpected. Drawing on repeating ideas and themes, summarize the findings in relation to your research question(s) and to previous research.

**Report results**

To report qualitative results, present repeating ideas that lead to major themes that, in turn, inform conclusions and implications. Conclusions are statements that interpret and evaluate the results found from the study. Make sure to give primary emphasis to the results that relate to the research questions of your study. Quote one or two responses that exemplify a repeating idea. Quotations, which capture the words, emotions, experiences, and perceptions of interviewees, are not easily dismissed by readers. You may also want to quote a response that was an exception to a trend in order to illustrate a minority opinion or highlight a noteworthy idea. If so, report that it is one person's response. Finally, make sure to discuss what practical or theoretical implications can be drawn for your findings, any major shortcomings or limitations of the methodology used, and directions or suggestions for future research.

### 3.2.4 Reporting experiment results

**Analyze the data**

Calculate descriptive statistics on outcome measures and determine if the variables are normally distributed, a requirement for many statistical tests. If a variable is not normally distributed, consult with a statistician to determine if you need to transform the variable. While comparing group means and standard deviations will give you a rough sense of group differences on outcome measures, you must use statistical tests to demonstrate that these differences are unlikely to have occurred by chance.

**Determine findings**

From your data analysis, summarize the findings in relation to your research question(s) or hypotheses and to previous research findings.

**Report results**

Conclusions are statements that interpret and evaluate the results found from the study. Make sure to give primary emphasis to the results that relate to the research questions of your study such as the effect of an intervention. One way to represent the magnitude of an intervention effect is with a percentage change. For example, you might report that scores increased 35% for an intervention group compared with 15% for a control group that did not receive the intervention. Also discuss what practical or theoretical implications can be drawn for your findings, any major shortcomings or limitations of the methodology used, and directions or suggestions for future research.

### 3.2.5 Quantitative content analysis findings

Determining the findings of your content analysis involves more than simply reporting initial results. Instead, it is important to critically examine results and check for statistical pitfalls to develop accurate findings upon which you can make reliable conclusions.

**Critically examine results**

- No matter what your results, ask some critical questions:
- Were the criteria you selected valid indicators of content quality? Did you omit important criteria or include unnecessary ones?
- If you implemented an intervention and are comparing content between/among groups or periods of time,
- were there significant differences between/among groups on the content before the intervention started?
- were conditions for groups roughly the same (for example, equivalent classrooms, instruction, and assistance outside of class)?
- did anything happen other than your instructional intervention that would have affected study results?
- was there any difference in motivation between/among groups before or during the study?

**Check for statistical pitfalls**

While any conclusive findings should be statistically significant, having statistically significant results does not mean, they are important or valuable; it just indicates that the difference you found is unlikely to be due to chance.

If you used multiple raters/coders, is the level of interrater reliability acceptable (e.g., .70 or higher)? Do results indicate any type of bias on the part of one or more of the raters/coders? If you find poor reliability or suspected bias, your results are possibly unreliable and data should be regathered and/or reanalyzed.

If you are comparing content between/among groups, could there be any errors due to sample size? If you have fewer than 25 cases per group, you may lack adequate statistical power to detect differences between groups. On the other hand, if you have very large groups, almost any difference, even a trivial one, will be statistically significant, and could lead you to make unwarranted conclusions. For this reason, you should indicate effect sizes,

which allow the readers to judge how meaningful the differences are between/among groups.

**Other statistical pitfalls**

Consult with a statistician if you are unable to resolve statistical problems on your own.

**Make conclusions**

- Evaluate your results based on how well they answer your research questions or confirm your hypotheses.
- Statistically significant causal, predictive, or correlational findings, as well as important qualitative findings, should form the basis of your main conclusions. Emphasize your strongest findings.
- If you are evaluating an intervention using content analysis, consider all possible explanations for results before concluding an intervention definitely worked or did not work.
- Verify (triangulate ) findings from your content analysis with results from other data sources such as interviews or surveys that can provide additional insight. Finding similar results using different methods strengthens conclusions. On the other hand, differing results call for further analysis.

**Definitions**

**Member checking**: Getting feedback from participants in the study to check the researchers' interpretation.

**Narrative analysis:.**

**Open-ended interviewing**: A form of interviewing that does not employ a fixed interview schedule, but allows the researcher to follow the respondent's lead by exploring topics in greater depth and also by pursuing unanticipated topics.

**Open-ended questions**: Interview or survey questions that are to be answered in the respondent's own words, rather than by selecting pre- formulated responses.

**Participant observation**: A form of observation in which the researcher participates in the activities going on in a natural setting and interacts with people in that setting, rather than simply recording their behavior as an outside observer.

**Qualitative research**: A strategy for empirical research that is conducted in natural settings, that uses data in the form of words (generally, though pictures, artifacts, and other non-

quantitative data may be used) rather than numbers, that inductively develops categories and hypotheses, and that seeks to understand the perspectives of the participants in the setting studied, the context of that setting, and the events and processes that are taking place there.

**Rich data**: Data that are detailed, comprehensive, and holistic.

**Robustness:** Interpretations, results, or data that can withstand a variety of validity threats because they hold up even if some of the underpinnings are removed or prove incorrect.

**Summative evaluation**: Evaluation that is aimed at assessing the value of a developed program for the purpose of administrative or policy decisions. This evaluation often is done by testing the impact of the program after it has been implemented. See formative evaluation.

**Triangulation:** The cross-checking of inferences by using multiple methods, sources, or forms of data for drawing conclusions.

**Validity:** The truth or correctness of one's descriptions, interpretations, or conclusions.

**Validity threat:** A way in which one's description, interpretation, or con- clusion might be invalid, also known as "rival hypothesis" or "alternative explanation."

**My notes**

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